# CV

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**Namhyun Kim** namhyunk@asu.edu

Website Google Scholar

## Research Interests

Integrated sensing and communication (ISAC); AI/ML-based communications; wireless foundation models; massive MIMO; beyond-5G/6G communication systems; semantic communications; Bayesian estimation.

## Education

Arizona State University (ASU), Tempe, AZ, USA — Jan. 2025 – Present
Doctor of Philosophy in **Electrical, Computer and Energy Engineering**
*Advisor:* Prof. Ahmed Alkhateeb.

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea — Mar. 2021 – Feb. 2023
Master of Science in **Electrical Engineering** — Cumulative GPA: 3.81/4.30
*Dissertation:* Location-Aware Downlink Beamforming with Marginal Training Overhead for MmWave FDD Massive MIMO Systems.
*Advisor:* Prof. Junil Choi.

Yonsei University, Seoul, South Korea — Mar. 2015 – Feb. 2021
Bachelor of Science in **Electrical Engineering** (High Honors) — Cumulative GPA: 4.04/4.30

## Professional Experience

**InterDigital, Inc.** — Philadelphia, PA, USA
*Ph.D. Intern, Wireless Lab* — June 2026 – Aug. 2026

- Calibrated integrated sensing and communication (ISAC) channel models based on 3GPP TR 38.901 for joint sensing and communication simulations.
- Developed a semantic communication framework incorporating 3GPP TR 38.901 spatial consistency procedures to capture channel evolution across user locations.

**SK Telecom, Co., Ltd.** — Seoul, South Korea
*Manager, LTE/5G Radio Access Network (RAN) Performance Improvement Group* — Jan. 2022 – Sep. 2023

- Performed LTE/5G L1/L2/L3 network planning, engineering, troubleshooting, and performance analysis.
- Conducted commercial verification testing of 64-TRX massive MIMO systems with Samsung Electronics, Co., Ltd.
- Used Python, SQL, Kubernetes, Linux, and MATLAB for network engineering and analysis.

## Publications

*Manuscripts under Revision and Preprints*

- **N. Kim**, S. Alikhani, and A. Alkhateeb, "LWM-TF: A Wireless Foundation Model on 5G NR Time-Frequency Resource Grids," *IEEE Journal of Selected Topics in Signal Processing (JSTSP)*, **under revision**. [arXiv].
- **N. Kim**, M. S. Abouamer, J. Park, P. Popovski, and A. Alkhateeb, "Digital Twin-Aided Prescreening for User Scheduling in MU-MIMO Downlink Systems," arXiv preprint arXiv:2608.10364, 2026. [arXiv].

*Journal Articles*

- **N. Kim**, J. Han, J. Choi, A. Alkhateeb, C.-B. Chae, and J. Park, "Integrated Sensing and Communications in Downlink FDD MIMO Without CSI Feedback," *IEEE Transactions on Wireless Communications*, 25:2984–3000, 2026. [Link].
- J. Han, **N. Kim**, and J. Park, "Reducing Latency by Eliminating CSIT Feedback: FDD Downlink MIMO Transmission for Internet-of-Things Communications," *IEEE Internet of Things Journal*, 13(6):11301–11315, 2026. [Link].
- **N. Kim**, I. P. Roberts, and J. Park, "Splitting Messages in the Dark—Rate-Splitting Multiple Access for FDD Massive MIMO Without CSI Feedback," *IEEE Transactions on Wireless Communications*, 24(4):3320–3332, 2025. [Link].

*Conference Papers*

- **N. Kim**, S. Alikhani, and A. Alkhateeb, "LWM-Spectro: A Foundation Model for Wireless Baseband Signal Spectrograms," *IEEE International Conference on Communications (ICC)*, Glasgow, UK, pp. 1–6, 2026. [Link].
- **N. Kim** and J. Park, "Robust Precoding and Rate-Splitting Multiple Access for FDD Massive MIMO Without CSI Feedback," *IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, MI, USA, pp. 1–6, 2025. [Link].
- **N. Kim**, J. Han, and J. Park, "Integrated Sensing and Communications in FDD MIMO Without CSI Feedback: Towards FDD MIMO ISAC," *22nd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, Seoul, South Korea, pp. 132–137, 2024. [Link].

## Projects

**LTE Link-Level Simulation** Seoul, South Korea
*Intelligent & Secure underwater CommunicaTion (InSeCT), KRIT* Sep. 2023 – Apr. 2024

- Implemented the signal processing chain in an OFDM receiver.
- Derived a post-equalization SINR model for CFO-impaired OFDM transmission.
- Linked post-equalization SINR to BICM capacity.

**5G Shared Network Commercial Deployment** Daejeon, South Korea
*SK Telecom* Sep. 2022 – Aug. 2023

- Expanded 5G coverage in Korea through shared-network deployment with KT and LGU.
- Integrated and optimized network equipment while addressing security concerns related to shared infrastructure.
- Managed traffic loads and coordinated RAN operations to maintain service quality.

**Undergraduate Capstone Project** Seoul, South Korea
*School of Electrical Engineering, Yonsei University* Mar. 2020 – Aug. 2020

- Completed a department-led capstone project during the COVID-19 pandemic.
- Developed a campus access control system combining RFID and temperature measurement sensors.

## Scholarships & Awards

**Global Korea Scholarship: Study Abroad Scholarship Program** NIIED, Korean Government
National scholarship supporting graduate studies abroad. 2025–2026

**Graduate Student Academic Excellence Scholarship** Wooyang Foundation
Merit-based scholarship recognizing academic excellence. 2021–2022

**Korean Government-funded Student** KAIST
Master's studies fully funded by Korean government scholarships. 2021–2022

**Department Chair Commendation** Dept. of EE, Yonsei University
Recognized for tutoring international students at Songdo International Campus. Feb. 2021

**Academic Excellence Award** Yonsei University
Semester Honors (Spring 2016, Fall 2018, Spring 2019), High Honors (Spring 2020), Highest Honors (Fall 2019).

**Graduation with Distinction** Yonsei University
Graduated with High Honors, placing in the top 3% of electrical engineering graduates. Feb. 2021

## Teaching and Mentoring Experience

**Graduate Teaching Assistant** Daejeon, South Korea
*Course: EE528: Engineering Random Processes, KAIST* Sep. 2021 – Feb. 2022

- Served as a teaching assistant for Prof. Junil Choi's Engineering Random Processes course.

**Academic Tutor for International Students** Songdo, Incheon, South Korea
*EE Honor Society Program, Dept. of Electrical Engineering, Yonsei University* Sep. 2019 – Feb. 2021

- Tutored 21 international students in Engineering Mathematics and Engineering Physics.
- Conducted tutoring sessions at students' request; most participants achieved strong grades.

## Skills

- **Programming:** Python, C/C++, MATLAB, SQL.
- **Machine Learning:** PyTorch; self-supervised pre-training and fine-tuning of transformer foundation models.
- **Wireless Simulation:** NVIDIA Sionna (5G NR PUSCH link-level simulation), 3GPP TR 38.901 channel modeling.
- **Software:** Linux, Kubernetes.
- **Collaboration:** Cross-cultural research and engineering across academia and industry in the U.S. and South Korea.